

INTERNSHIP VIVA COMPLETE NOTES

1. Introduction

I worked on a College Management System backend using Node.js and Express. I developed APIs for login, registration, student, faculty and admin modules. I also implemented database integration, CRUD operations and authentication.

2. Architecture Flow

Client → Route → Controller → Service → Database → Response

3. Concepts

Node.js: JavaScript runtime for backend

Express.js: Framework for routing and APIs

API: Communication between frontend and backend

CRUD: Create, Read, Update, Delete

4. Layers

Route: Handles request

Controller: Handles request/response

Service: Business logic

Model: Database structure

5. Important Questions

Route kyu? → request handle karne ke liye

Controller kyu? → logic separate rakhne ke liye

Service kyu? → clean & reusable code

Validation kyu? → wrong data rokne ke liye

try-catch kyu? → error handle karne ke liye

6. API Methods

GET → fetch

POST → insert

PUT → update

DELETE → delete

7. Status Codes

200 → success

400 → bad request

500 → server error

8. Authentication

User login karta hai, backend verify karta hai aur access deta hai

9. Async/Await

Asynchronous code ko readable banane ke liye use hota hai

10. Challenges

Backend samajhna, debugging, database issues — solved with practice

11. Final Line

I followed layered architecture to keep code scalable and maintainable.